Score Normalization and Vote Construction Govern Preference-Graph Repair Outcomes in Multi-Ranker Retrieval

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Preference-graph repair is often motivated by the idea that removing cyclic inconsistency should improve downstream ranking quality, but that conclusion depends on how the graph is constructed. We study this question on SciDocs, FiQA, HotpotQA, and BRIGHT using stored BM25, TF-IDF, and MiniLM candidate scores, fixed candidate pools, and public qrels. Our primary analysis uses a normalized retention-matched protocol: per-query, per-ranker min–max normalization followed by retention-matched vote and edge thresholds. As a raw-score ablation, we evaluate the unnormalized construction that treats native heterogeneous score margins as directly comparable. That unnormalized construction is severely scale dominated: when BM25 participated in an edge, its conditional mean weight share was 0.988, versus 0.512 under the primary protocol. Normalization and vote construction materially changed retained votes, edge weights, cyclicity, removed edges, and several repaired-versus-unrepaired signs. Under the primary protocol, repair still changes graph structure, but structural consistency and retrieval effectiveness remain distinct: repair can reduce graph inconsistency without producing a reliable improvement in ranked retrieval. No repaired-versus- unrepaired improvement remains robust after permutation testing, bootstrap uncertainty, influence analysis, and multiplicity correction. Simple baselines such as reciprocal rank fusion and CombSUM also remain strong comparators under the normalized graphs. The conclusion is therefore methodological as well as empirical: score normalization and vote construction jointly determine the preference graph on which repair operates, and claims about repair effectiveness must report those choices explicitly rather than treating graph construction as a neutral preprocessing step.

CCS Concepts: • **Information systems → Data cleaning**; **Data quality**; **Retrieval models and ranking**.

Additional Key Words and Phrases: preference graphs, data quality, structural inconsistency, rank aggregation, feedback arc set repair, retrieval evaluation, benchmark curation

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1 INTRODUCTION

Multi-ranker retrieval pipelines combine sparse retrievers, dense retrievers, and other learned ranking components whose outputs disagree with one another as a matter of course, and that disagreement must be reconciled rather than trusted from a single source [1, 5, 8, 16]. Preference graphs are one way to keep that disagreement visible instead of collapsing it prematurely: for a given query, a directed edge from document u to document v records evidence that u should outrank v, and the edge weight records how strongly that evidence should count. The representation is appealing precisely because it defers the fusion decision — it makes conflicting evidence explicit graph structure rather than a single scalar score computed once and discarded. But that deferral only pays off if the graph itself is trustworthy. A preference graph built from heterogeneous ranker scores is a *derived data artifact*: its edges, weights, and cycles are only as meaningful as the score-comparability assumptions used to build it. If native BM25, TF-IDF, and neural similarity margins are treated as directly comparable evidence without justification, the resulting graph can encode score scale rather than evidential strength, and every downstream measurement — cyclicity, repair, and retrieval effectiveness alike — inherits that distortion silently.

That issue matters because preference-graph repair operates entirely on the constructed graph. Once weighted pairwise evidence is aggregated, the graph may contain directed cycles, so no single ranking can satisfy every recorded preference. A standard response is to repair the graph by removing a minimum-weight feedback arc set before extracting a final ranking [1, 14]. This intervention improves structural consistency relative to the graph's own edge weights, but structural consistency and retrieval effectiveness are not the same objective. A repair step can remove cycle-inducing edges, reduce a graph-internal inconsistency measure, and still fail to improve a relevance metric such as normalized discounted cumulative gain (nDCG) if the modified edges do not alter the final ranking in a helpful way. For that reason, a study of preference-graph repair must examine not only the repair algorithm but also the graph-construction choices that determine what the repair sees.

This paper studies that upstream dependency directly. Existing rank-aggregation, feedback-arc-set, and fusion literatures largely study aggregation algorithms, repair algorithms, or fusion functions on the assumption that the input preference signal is already well formed; the empirical quality of the preference graph itself is treated as a solved or irrelevant precondition rather than as a variable to be measured. This paper instead studies the quality, robustness, and downstream consequences of preference-graph *construction*: rather than asking for a predictive answer to "when repair helps," we treat score normalization and vote construction as first-class experimental variables and examine how those choices govern the preference graph on which repair operates, and how they affect both structural outcomes and downstream retrieval behavior. We evaluate four benchmarks—SciDocs, FiQA, HotpotQA, and BRIGHT—with stored BM25, TF-IDF, and MiniLM score files [6, 18, 23, 28]. Our primary protocol applies per-query, per-ranker min–max normalization followed by retention-matched thresholding; we retain the unnormalized raw-margin construction only as an ablation. The motivation for this comparison is empirical, not cosmetic. Under the raw-margin construction, BM25 contributed nearly all aggregate edge weight whenever it participated, so the graph was largely BM25-weighted even when three rankers were nominally present. Treating such raw heterogeneous margins as directly comparable support signals is not a neutral preprocessing choice; it is a substantive graph-construction assumption that shapes every downstream result.

Once graph construction is treated as a first-class variable, several measured outcomes change. Normalization and vote construction materially alter retained votes, edge weights, cyclicity, removed edges, and some repaired-versus-unrepaired signs. They also sharpen the distinction between total cyclicity and the smaller amount of nontrivial cyclicity that remains after direct mutual pairs are deleted. Repair therefore remains structurally active under the normalized graphs, but its retrieval effects are method-dependent and construction-sensitive rather than stable properties of "repair" in the abstract. The central empirical result is not that one dataset provides a robust positive exception, nor that repair has a construction-independent null effect. It is that graph-construction choices substantially determine both the structure being repaired and the retrieval conclusions drawn from that repair.

This framing changes how the paper's contributions should be understood. We do not present feedback-arc-set repair as a generally superior reranking strategy, we do not propose a new feedback-arc-set solver or fusion rule, and we do not interpret raw-margin results as primary evidence. Instead, the paper makes the following contributions:

- **A reproducible, normalization-explicit construction protocol.** We formalize per-query, per-ranker score normalization and retention-matched vote thresholding as reportable, first-class design choices for multi-ranker preference graphs, rather than leaving them as implicit preprocessing (Section 3).
- **A systematic empirical audit of preference-graph quality.** Across four benchmarks and three vote-extraction regimes, we measure how normalization changes retained votes, edge weights, cyclicity, and removed edges, and we separate direct mutual contradictions from longer-cycle nontransitivity (Section 5).
- **Evidence separating structural consistency from retrieval effectiveness.** We show that feedback-arc-set repair reduces graph-internal inconsistency in the active regimes while producing no repaired-versus-unrepaired retrieval effect that survives paired testing, influence analysis, and multiplicity correction (Section 6).
- **A raw-versus-normalized sensitivity analysis.** We show that several repaired-versus-unrepaired signs, and the graph's dependence on individual rankers, are not stable under defensible alternatives to raw-margin construction, demonstrating that graph construction is not a neutral preprocessing step (Sections 5–6).
- **Rigorous statistical validation.** Every repaired-versus- unrepaired claim is evaluated under paired permutation testing, bootstrap uncertainty, leave-one-out influence analysis, and Holm/Benjamini–Hochberg multiplicity correction, rather than reported as an unadjusted point estimate (Section 4.5).

These contributions support a more limited but more defensible conclusion: preference-graph repair outcomes are not invariant to score normalization and vote construction, and structural consistency should not be assumed to translate into retrieval improvement without direct evaluation.

The remainder of the paper is organized as follows. Section 2 positions the study relative to data-quality frameworks, rank aggregation, and rank fusion. Section 3 formalizes preference graphs, normalized vote construction, structural metrics, and feedback-arc-set repair. Section 4 describes the four-dataset protocol, vote-extraction regimes, baselines, and statistical evaluation policy. The results sections then report normalized structural behavior, repaired-versus-unrepaired retrieval effects, raw-versus-normalized ablations, and the implications of those findings for baseline comparison and practical use. The final sections discuss limitations, reproducibility, and the broader lesson that graph-construction choices must be reported explicitly in any empirical claim about preference-graph repair.

2 BACKGROUND AND RELATED WORK

Preference-graph methods sit at the intersection of several literatures: rank aggregation, noisy pairwise-comparison ranking, retrieval fusion, and data quality for derived analytical artifacts. The goal of this paper is narrower than a new ranking-method contribution. We do not introduce a new feedback-arc-set solver, a new retrieval ranker, or a new fusion rule. Instead, we study how heterogeneous ranker scores are converted into weighted preference graphs and how those graph-construction choices govern the measured effect of repair on both structural consistency and downstream retrieval.

Rank aggregation and preference graphs. Preference graphs are a standard representation for aggregating pairwise evidence across voters, rankers, or judges [1, 8, 20]. For a query-specific candidate set, a directed edge $u \rightarrow v$ records support for ranking document u ahead of document v, while the edge weight records the strength of that support. This representation is useful because it preserves disagreement as explicit graph structure instead of collapsing it immediately into a single fused score.

Feedback arc set repair and Kemeny-style consistency. Once a preference graph contains a directed cycle, no total ranking can satisfy every recorded preference simultaneously. Kemeny's classical consensus ranking minimizes total pairwise disagreement with input preferences [13]; deciding an optimal Kemeny ranking and computing a minimum feedback arc set are closely related combinatorial problems [1, 8]. Feedback arc set is NP-hard [12], and the weighted formulation considered here generalizes that problem, so algorithms either optimize exact formulations on moderate instances or use heuristics. Classical work therefore studies minimum weighted feedback arc set (MW-FAS) and Kemeny-style objectives that remove or violate as little edge weight as possible to obtain an acyclic ordering [1, 8, 10, 14]. That optimization target is purely graph-internal. It formalizes *structural* consistency, not relevance to external retrieval judgments. This distinction is central here: a better solution to the graph's own repair objective need not improve a metric such as nDCG.

Probabilistic ranking from noisy pairwise preferences. Classical Bradley–Terry models treat pairwise outcomes as noisy comparisons about latent item strengths [3]. The Plackett–Luce model generalizes this pairwise formulation to full rankings by factorizing the probability of an observed permutation into a sequence of Bradley–Terry-style sequential choices [17, 21]. We do not fit a Plackett–Luce model in our experiments, but its presence in this family clarifies that Rank Centrality and Bradley–Terry-style comparators are instances of a broader probabilistic-ranking literature that infers a latent ordering from pairwise or listwise outcomes, as distinct from the graph-repair objective studied here. Rank Centrality likewise models pairwise outcomes as noisy evidence about a latent ordering and derives node scores from the induced comparison graph [20]. Those probabilistic methods are useful here as graph-dependent baselines; graph-repair methods instead remove edges to enforce acyclicity rather than fitting a probabilistic ranking model. Both families share the same upstream dependence: the object they consume is the constructed preference graph, so changing vote extraction or edge weighting can alter their behavior even when the upstream retrieval runs are unchanged.

Rank fusion in information retrieval. Information-retrieval systems already combine heterogeneous signals through score or rank fusion, including reciprocal rank fusion (RRF) and Borda-style methods [7], and CombSUM and CombMNZ, the summation- and vote-weighted score-combination rules introduced by Fox and Shaw [11] and analyzed further by Lee [15]. That analysis clarifies how score- and rank-based fusion behave when several retrieval strategies contribute overlapping evidence [15]. Fusion methods are not uniformly beneficial: Benham and Culpepper show that rank fusion carries a query-level risk-reward trade-off, improving some queries while harming others [2]. The helped/harmed/unchanged query counts we report for repair (Section 6) document an analogous risk profile for graph repair, rather than a uniform improvement, so this literature's caution about fusion generalizes to graph-based reranking as well. These baselines matter here for two reasons. First, they provide graph-free comparators against which repaired and unrepaired graph methods must be judged. Second, they make explicit a design choice that preference-graph work can hide: when heterogeneous retrieval scores are merged, some form of scale handling or an explicit decision to ignore cross-model score comparability becomes part of the method before score magnitudes are interpreted jointly. We keep CombSUM as the score-summation baseline and do not expand the primary baseline family with CombMNZ (an exploratory stored-score CombMNZ check did not overturn CombSUM's role enough to justify expanding the baseline table); the study's comparative focus is repair sensitivity under fixed pools rather than an exhaustive fusion bake-off.

Score normalization for heterogeneous retrieval signals. BM25, TF-IDF, and neural similarity models operate on different native score scales. A raw difference of 0.05 has no reason to represent comparable evidential strength across those models, a point long recognized in meta-search score-normalization work [19]. This scale-comparability problem is not unique to preference graphs: recent analysis of fusion functions for hybrid lexical-semantic retrieval shows that convex-combination fusion of BM25 and dense-retrieval scores is highly sensitive to how those scores are normalized before combination, and that reciprocal-rank fusion is attractive in part because it sidesteps the normalization decision entirely by discarding score magnitude [4]. The methodological issue studied in this paper is therefore not whether preference graphs are useful in principle, but whether weighted graph construction is stable when native heterogeneous score margins are treated as if they were directly comparable. Our primary normalized protocol makes that assumption explicit and testable rather than leaving it implicit in the edge weights.

LLM pairwise ranking and inconsistency. Recent work has used large language models as retrieval judges, rerankers, and pairwise preference sources [22, 24]. That line of work makes inconsistency salient, but it does not remove the measurement problem studied here. Whether the pairwise evidence comes from classical rankers or from LLM judgments, downstream claims about "repair" still depend on how pairwise evidence is filtered, weighted, and aggregated into a graph.

Derived-artifact data quality. We treat the preference graph itself as a derived data artifact whose properties can be audited. Cyclicity, strongly connected components, mutual contradictions, and removed feedback-arc weight are intrinsic properties of the graph produced by a construction pipeline, whereas retrieval metrics such as nDCG are extrinsic properties of rankings evaluated against qrels. This split echoes a distinction long established in the data-quality literature between intrinsic data quality, which a data artifact has independent of any particular use, and contextual data quality, which depends on the task the data supports [27]: our intrinsic graph diagnostics (Section 3.7) describe the constructed graph on its own terms, while nDCG and the qrels-anchored diagnostics describe its fitness for the downstream retrieval task, and the central empirical finding of this paper is that these two families of measurement can move independently of one another. The contribution of this paper is to separate those two measurement families and show that conclusions about repair are not invariant to score normalization and vote construction. In this sense, the paper is a study of graph construction and measurement sensitivity, not a claim that a particular repair heuristic solves retrieval reranking.

3 METHODOLOGY

This section defines the normalized preference-graph pipeline used for the primary analysis. The order matters. Upstream score scale affects pairwise margins; pairwise margins affect retained votes; retained votes determine weighted graph edges; graph structure determines what repair sees; and repair influences ranking extraction only through the modified graph. Table 1 summarizes notation used in this section and Section 4, and Figure 1 summarizes the end-to-end pipeline these subsections define, stage by stage.

3.1 Upstream Scores and Candidate Pool

The primary pipeline starts from stored score files for three upstream rankers: BM25, TF-IDF, and MiniLM. For each query q, we first define a shared candidate pool D_q from the union of documents scored by at least one ranker. If that union already has size at most the dataset-specific pool size, we keep all documents. Otherwise we rank the union by reciprocal rank fusion with parameter $k = 60$ over the full per-ranker native ranked lists and keep the top-$|D_q|$ candidates. Candidate pooling therefore uses an RRF-family computation, but it is distinct from both the Prior ranking and the reported RRF baseline because it is applied before graph construction and before candidate restriction.

Table 1. Notation used in Sections 3 and 4.

Symbol	Meaning
q, D_q	Query and its candidate document set
s	Upstream ranker in {BM25, TF-IDF, MiniLM}
$x_{q,s}(d)$	Native score assigned by ranker s to document d for query q
$\bar{x}_{q,s}(d)$	Normalized per-query, per-ranker score
$m_{q,s}(u,v)$	Absolute pairwise margin for pair $\{u,v\}$ under ranker s
$G_q = (V_q, E_q, w_q)$	Preference graph for query q ($V_q = D_q$)
$\tilde{G}_q$	Repaired (acyclic) preference graph
r	Vote-extraction regime, $r \in \{\texttt{ms2}, \texttt{ms1}, \texttt{ms1\_drop\_mutual}\}$
κ_r	Regime-specific minimum support threshold
λ_s	Per-ranker vote-retention threshold within one dataset/protocol/regime cell
γ_r	Aggregate edge-retention threshold within one dataset/protocol/regime cell
$c_q(u,v)$	Support count for direction $u \rightarrow v$
$a_q(u,v)$	Aggregate retained weight for direction $u \rightarrow v$
F_{opt}	Exact minimum-weight feedback arc set (optimization target)
F_{greedy}	Greedy removed-edge set computed in the experiments
$\deg^+(u), \deg^-(u)$	Out-degree, in-degree of node u in a graph
$s_{\text{cop}}, s_{\text{bal}}, s_{\text{mc}}$	Copeland, weighted balance, and Markov graph scores
$\hat{s}(\cdot)$	Min–max normalization of a score vector to $[0,1]$
α	Hybrid mixing weight for the graph-derived component
π	A ranking (permutation) of V_q
$\text{BEW}(\pi, G)$	Backward-edge weight of π relative to graph G
$\text{PIC}(\pi, G)$	Pairwise inconsistency count (unweighted analogue)
k	nDCG cutoff

3.2 Per-Query, Per-Ranker Score Normalization

We apply per-query, per-ranker min–max normalization. The transformed scores are not probabilistically calibrated probabilities: they are ordinal, per-query, per-ranker rescalings to $[0,1]$, not the outcome-frequency estimates produced by Platt scaling or isotonic regression, and no such probabilistic claim is made or implied anywhere in this paper.

The primary protocol does not treat native BM25, TF-IDF, and MiniLM margins as directly comparable edge-weight contributions. For each query q and ranker s, let $D_{q,s} \subseteq D_q$ be the candidate documents for which ranker s supplies a score. We define the normalized score

$$\bar{x}_{q,s}(d) = \begin{cases} 0, & \max_{d' \in D_{q,s}} x_{q,s}(d') - \min_{d' \in D_{q,s}} x_{q,s}(d') \leq 10^{-12}, \\ \dfrac{x_{q,s}(d) - \min_{d' \in D_{q,s}} x_{q,s}(d')}{\max_{d' \in D_{q,s}} x_{q,s}(d') - \min_{d' \in D_{q,s}} x_{q,s}(d')}, & \text{otherwise.} \end{cases} \tag{1}$$

This transformation is applied separately for each query and ranker after candidate restriction. Missing documents remain missing: if a ranker did not score document d, no normalized score is imputed for that document. If a ranker's candidate-restricted score vector is constant, every normalized score is set to 0 and that ranker contributes no nonzero within-query margin.

Normalization is necessary because the edge weights in a multi-ranker preference graph inherit whatever comparability assumptions are imposed on the upstream scores. We therefore treat the unnormalized raw-margin construction only as an ablation and use Eq. (1) as the primary graph-construction input.

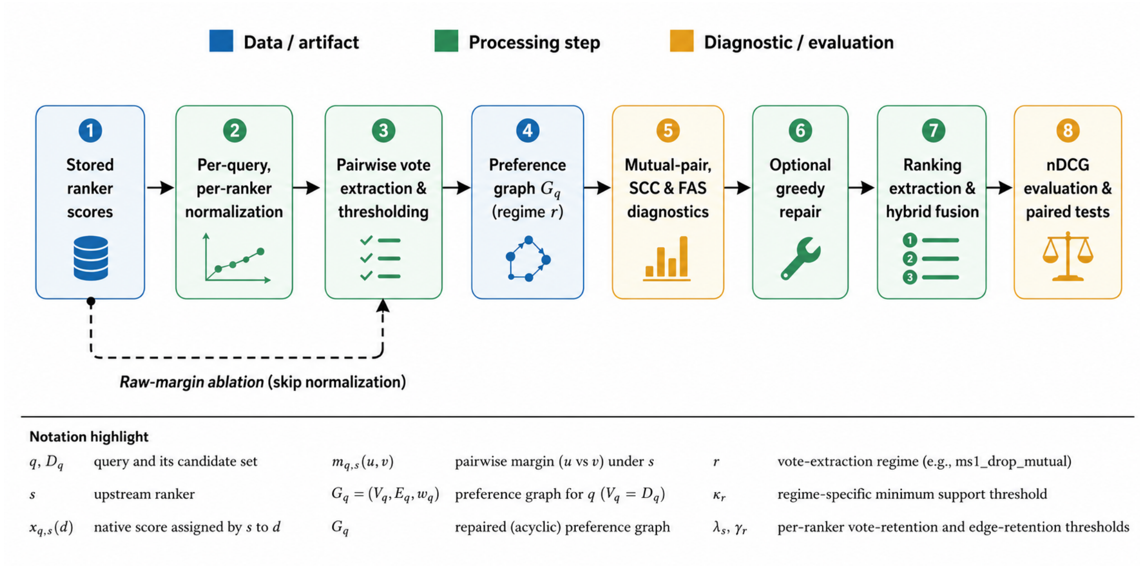

Fig. 1. The eight-stage evaluation pipeline (numbered 1–8), color-coded by role: blue stages are data artifacts, green stages are processing steps, and amber stages are diagnostic or evaluation steps. Stored ranker scores are normalized per query and per ranker (Eq. (1)), converted to pairwise votes and thresholded into a preference graph G_q under regime r, diagnosed structurally, optionally repaired, and finally converted to a ranking and scored by nDCG under paired statistical tests. The dashed arrow marks the raw-margin ablation, which reuses every downstream stage unchanged but skips normalization and keeps fixed raw-margin thresholds. A notation-highlight key below the diagram repeats the symbols most relevant to this pipeline from Table 1. The reader should take away that normalization and repair are two distinct, separated stages, not one combined step.

3.3 Pairwise Vote Extraction and Graph Construction

For each query q, ranker s, and unordered candidate pair $\{u, v\} \subseteq D_q$, the primary protocol compares normalized scores only if both documents are present in $D_{q,s}$. If either document is missing, ranker s abstains on that pair. If $\tilde{x}_{q,s}(u) = \tilde{x}_{q,s}(v)$, ranker s also abstains. Otherwise the absolute pairwise margin is

$$m_{q,s}(u, v) = \left|\tilde{x}_{q,s}(u) - \tilde{x}_{q,s}(v)\right|. \tag{2}$$

Ranker s casts a directional vote only when $m_{q,s}(u, v) \geq \lambda_s$, where λ_s is the per-ranker vote-retention threshold for the current dataset, protocol, and regime. The direction is given by the larger normalized score. For a candidate direction $u \longrightarrow v$, let

$$c_q(u, v) = \sum_s \mathbf{1}[s \text{ retains } u \longrightarrow v] \tag{3}$$

be the support count and

$$a_q(u, v) = \sum_s m_{q,s}(u, v)\, \mathbf{1}[s \text{ retains } u \longrightarrow v] \tag{4}$$

be the aggregate directional weight. A direction is retained only if

$$\begin{aligned} c_q(u, v) &\geq \kappa_r, \\ a_q(u, v) &\geq \gamma_r, \end{aligned} \tag{5}$$

where κ_r is the regime-specific minimum support and γ_r is the aggregate threshold for the current dataset/protocol/regime cell. The graph builder then sums all retained per-ranker contributions for the same directed pair, so the final edge weight is $w_q(u,v) = a_q(u,v)$ whenever Eq. (5) is satisfied. Because each direction is tested independently, a single unordered pair can retain both $u \to v$ and $v \to u$.

3.4 Vote-Extraction Regimes

We evaluate three vote-extraction regimes over the same stored score files and candidate pools:

- **`ms2`** requires support $\kappa_r = 2$ and applies a nontrivial aggregate threshold γ_r. Under the unnormalized raw ablation this threshold is 0.1; under the primary protocol it is retention-matched separately for each dataset.
- **`ms1`** requires support $\kappa_r = 1$ and uses the dataset-specific aggregate threshold for that protocol/regime cell. It preserves directional disagreement whenever both directions independently survive thresholding.
- **`ms1_drop_mutual`** uses the same support rule as `ms1`, but after thresholding deletes every unordered pair that retained both directions. It therefore treats direct bidirectional contradiction as unresolved rather than retaining either arc.

The interpretation of these regimes is important. Cyclicity under `ms1` does not automatically imply longer-cycle nontransitivity; it can arise from direct mutual contradictions alone. We therefore separate mutual pair diagnostics from residual cyclic structure after mutual-pair deletion.

3.5 Retention-Matched Thresholds

Under the primary protocol, thresholds are not fixed globally at the raw values 0.05 and 0.1. Instead, for each dataset, regime, and normalization variant, we match retention behavior to the unnormalized raw-margin construction without using qrels. Let $\rho^{\text{raw}}_{s,r}$ denote the raw retained vote rate for ranker s in regime r, computed over the stored query subset with denominator equal to the number of candidate pairs for which both documents are scored by s and the native scores are unequal. For a normalization variant, we pool all normalized margins $m_{q,s}(u,v)$ across the dataset and set

$$\lambda_s = Q_{1-\rho^{\text{raw}}_{s,r}}\left(\left\{m_{q,s}(u,v)\right\}\right), \tag{6}$$

where Q_p denotes the empirical quantile at level p. If a ranker has no eligible margins in that cell, we set $\lambda_s = 0$.

After vote-threshold matching, we choose the aggregate threshold γ_r deterministically to minimize the absolute gap between the retained edge count under the normalized construction and the retained edge count under the raw reference construction for the same dataset and regime. The search is over the realized aggregate candidate weights in that cell; when multiple thresholds yield the same minimum gap, the smaller threshold is selected. Qrels are not used at any stage of this threshold selection procedure. This matching strategy keeps graph density and vote-retention behavior approximately comparable across normalization variants without tuning for nDCG.

Retention matching is an experimental control, not a claim about optimal threshold selection. Its sole purpose is to let a raw-versus-normalized comparison isolate the effect of score-scale handling: if thresholds were left unmatched, any structural or retrieval difference between the raw and normalized protocols could be confounded by unrelated differences in graph density or edge count rather than by score-scale handling itself. This procedure is therefore neither a universally optimal threshold-selection method nor a qrels-tuned one, and we do not present it as either. A sensitivity sweep of nearby retention/threshold policies (Section 9) shows that graph density, cyclicity, and removed-edge patterns can change, and that some repaired-versus-unrepaired cell signs flip, while no positive cell survives Holm correction under those alternatives. We therefore retain raw-reference retention matching as the primary protocol rather than treating any single nearby target as universally optimal.

3.6 Mutual Pairs and Nontrivial Cycles

We distinguish two kinds of cyclic structure. A *mutual pair* is a direct contradiction in which both $u \to v$ and $v \to u$ are retained. A *nontrivial cycle* is any directed cycle that remains after deleting all mutual-pair edges. The primary results therefore report cyclic query percentage and largest strongly connected component (SCC) size both before and after mutual-pair deletion.

3.7 Structural Diagnostics

We group structural measurements into intrinsic graph diagnostics and qrels-anchored diagnostics.

Intrinsic graph diagnostics. For each query graph we record edge count, graph density $|E_q|/(|D_q|(|D_q|-1))$, mutual-pair count, the share of queries containing any mutual pair, cyclic-query indicators before and after mutual-pair deletion, largest SCC size before and after mutual-pair deletion, directed triangle counts, total graph weight $W_q = \sum_{(u,v)\in E_q} w_q(u,v)$, and normalized graph weight $W_q/\max(1,|E_q|)$. After repair we record the number of removed edges, their total removed weight, and normalized removed weight W_q^{rem}/W_q when $W_q > 0$ and 0 otherwise.

Qrels-anchored diagnostics. We additionally report two diagnostics that compare a graph to an external reference ranking π_{ref} derived from relevance judgments: the **backward-edge weight**

$$\mathrm{BEW}(\pi_{\mathrm{ref}}, G) = \sum_{(u,v)\in E:\ \pi_{\mathrm{ref}}(v)<\pi_{\mathrm{ref}}(u)} w(u,v), \tag{7}$$

and the **pairwise inconsistency count**

$$\mathrm{PIC}(\pi_{\mathrm{ref}}, G) = \left|\{(u,v)\in E : \pi_{\mathrm{ref}}(v) < \pi_{\mathrm{ref}}(u)\}\right|, \tag{8}$$

the unweighted count of the same contradictions. These are not intrinsic graph properties and are not independent validation signals: they use the same qrels family as downstream retrieval evaluation. We therefore interpret them as reference-alignment diagnostics rather than as separate evidence of retrieval quality.

3.8 Graph Repair

The weighted feedback arc set optimization problem for graph G_q is

$$F_{\mathrm{opt}} = \operatorname*{arg\,min}_{F\subseteq E_q:\ (V_q, E_q\setminus F)\ \text{acyclic}} \sum_{(u,v)\in F} w_q(u,v), \tag{9}$$

with repaired graph $\widetilde{G}_q = (V_q, E_q \setminus F, w_q)$ when a set F is removed. The optimizer need not be unique; when multiple minimum-weight removal sets exist, any one optimal set is valid for the objective. In the primary experiments we do *not* solve Eq. (9) exactly. Instead, we compute a greedy removal set F_{greedy} by repeatedly finding a directed cycle in the current graph, removing the minimum-weight edge from that discovered cycle, and continuing until no cycle remains. The implementation uses `networkx.find_cycle` to identify a cycle at each step. This greedy cycle-peeling procedure is related in spirit to classical feedback-arc-set heuristics that remove edges based on local cycle structure [9], but it is a simple cycle-peeling variant rather than a reimplementation of that specific algorithm, and we claim no worst-case approximation bound for it.

This procedure approximately minimizes removed edge weight through greedy cycle peeling, but it has no approximation guarantee and its output can depend on cycle-discovery order. We therefore report removed weight and removed-edge overlap empirically, without labeling the computed solution as an exact minimum-weight feedback arc set. Repair remains a graph-internal operation: it depends only on w_q, not on qrels or nDCG.

For the exact robustness check in Section 4.4, we solve Eq. (9) with a linear-ordering MIP. For distinct nodes $u, v \in V_q$, let $x_{uv} \in \{0, 1\}$ mean that u precedes v in the recovered order. The program uses

$$x_{uv} + x_{vu} = 1 \qquad \text{for all unordered pairs } \{u, v\}, \tag{10}$$

$$x_{ab} + x_{bc} + x_{ca} \leq 2 \qquad \text{for all distinct triples } a, b, c, \tag{11}$$

$$\min \sum_{(u,v) \in E_q} w_q(u, v)\, x_{vu}. \tag{12}$$

An edge (u, v) is treated as reversed—and therefore removed—when $x_{vu} = 1$. The reported exact comparison uses SCIP via PySCIPOpt 6.2.1 with MIP gap 0, accepts only proven-optimal status, and applies a 300 s per-query safety limit that was never reached on any of the 1,025 query-regime graphs in this check (all solved to proven optimality; mean solve time 7.4 ms). Section 4.4 explains why this is one fewer than the $342 \times 3 = 1{,}026$ query-regime graphs implied by Table 2's usable-query counts. The 87.9% disagreement rate and 26.3% lower removed weight are computed on the 379 cyclic queries only, comparing removed-edge sets and total removed weight between exact and greedy repairs. No graph was excluded for timeout or non-optimal status.

3.9 Ranking Extraction

Ranking extraction is applied separately to G_q and $\widetilde{G}_q$. Graph construction and hybrid normalization must be kept distinct: the per-query, per-ranker score normalization in Eq. (1) is applied before graph construction, whereas the separate min–max normalization in Eq. (15) is applied only when combining graph-derived scores with a prior ranking.

We evaluate three graph-derived ranking families. The **Copeland score** is out-degree minus in-degree,

$$s_{\mathrm{cop}}(u) \;=\; \deg^{+}(u) - \deg^{-}(u), \tag{13}$$

counting net wins irrespective of edge weight. The **weighted balance score** is the signed sum of incident edge weight,

$$s_{\mathrm{bal}}(u) \;=\; \sum_{(u,v) \in E} w(u, v) \;-\; \sum_{(v,u) \in E} w(v, u). \tag{14}$$

The **Markov score** s_{mc} is the stationary distribution of a Rank-Centrality-style Markov chain on the weighted preference graph [20]. We use teleportation probability 0.15 (equivalently, retention probability 0.85), which yields an irreducible and aperiodic chain and therefore a unique stationary distribution even when the underlying preference graph is cyclic or disconnected. Final rankings sort by descending score with deterministic tie-breaking by document id when scores are equal; the Markov ranking additionally breaks ties by smaller weighted in-degree before document id.

The Prior ranking is a candidate-restricted RRF score over the stored ranker lists with parameter $k = 60$. The reported RRF baseline also uses $k = 60$ on the candidate-restricted lists, but its tie-breaking is not identical to Prior. We therefore name those roles separately rather than treating Prior and RRF as the same object.

The primary hybrid methods combine Prior with either Copeland or balance. Writing $\hat{s}(\cdot)$ for min–max normalization of a score vector to $[0,1]$ over the candidate set, the hybrid score is

$$s_{\text{hybrid}}(u) \;=\; \hat{s}_{\text{prior}}(u) \;+\; \alpha\,\hat{s}_{\text{comp}}(u), \qquad \text{comp} \in \{\text{cop}, \text{bal}\}, \tag{15}$$

with $\alpha = 0.3$ in the main comparisons. The primary analysis additionally reports a small alpha sensitivity sweep over $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$ for this primary protocol only.

3.10 Retrieval Evaluation

Retrieval quality is measured by nDCG at the dataset-specific cutoff used in candidate pooling. For a ranking π and a relevance map $\text{rel} : D_q \to \mathbb{Z}_{\geq 0}$,

$$\text{nDCG@}k(\pi) \;=\; \frac{\displaystyle\sum_{i=1}^{k} \frac{2^{\text{rel}(\pi_i)} - 1}{\log_2(i+1)}}{\displaystyle\sum_{i=1}^{k} \frac{2^{\text{rel}(\pi_i^{\text{ideal}})} - 1}{\log_2(i+1)}}, \tag{16}$$

where π_i is the document at rank i in π and π^{ideal} sorts candidate documents by descending relevance. nDCG@k is defined as 0 when the denominator is 0. All paired comparisons use only the intersection of queries for which both methods produce valid outputs.

3.11 Raw-Margin Ablation

The unnormalized raw-margin protocol is retained only as an explicitly labeled ablation. It uses native BM25, TF-IDF, and MiniLM scores without cross-ranker normalization; a fixed per-vote threshold of 0.05 for all three rankers; and fixed aggregate/support settings: `ms2` uses minimum support 2 and aggregate threshold 0.1, while `ms1` and `ms1_drop_mutual` use minimum support 1 and aggregate threshold 0.0. We do not interpret those raw margins as comparable confidence weights.

4 EXPERIMENTAL SETUP

4.1 Datasets and Stored Query Sets

We use four public retrieval benchmarks spanning distinct domains: SciDocs (scientific document retrieval) [6], FiQA (financial opinion and question answering) [18], HotpotQA (multi-hop question answering) [28], and BRIGHT (reasoning-intensive retrieval) [23]. SciDocs, FiQA, and HotpotQA are also part of BEIR [25]. The normalized-protocol rerun does not draw a fresh query sample. It reuses the fixed stored `query_ids.txt` files from the primary experiment inputs and then filters them only for evaluation eligibility.

A query is eligible if its qrels support ranking comparison: either at least two judged documents with at least two distinct relevance grades, or at least one judged document with strictly positive relevance. Queries failing this criterion are excluded before normalization, graph construction, and evaluation. The filter uses only qrels; it does not depend on method rankings or repair outcomes, and the same eligible query set is used for every compared protocol and method. On SciDocs, FiQA, and BRIGHT every stored query is eligible. On HotpotQA, 18 of the 70 stored IDs have only nonpositive relevance labels under the processed qrels and are therefore excluded, leaving 52 usable queries. Table 2 reports the stored query subset size, the usable query count under the primary protocol, the upstream retrieval depth used for stored score generation, the candidate-pool size, and the nDCG cutoff.

Table 2. Stored query subsets and evaluation settings reused by the four-dataset primary analysis.

Dataset	Stored IDs	Usable queries	Top-n	$\lvert D_q \rvert$	nDCG@k
SciDocs	120	120	50	20	20
FiQA	120	120	50	20	20
HotpotQA	70	52	35	10	10
BRIGHT	50	50	50	20	20

4.2 Upstream Rankers and Candidate Pooling

The upstream score files are fixed across all protocols and regimes. They come from BM25, TF-IDF, and MiniLM and are never regenerated in the normalized-protocol study. Candidate pooling uses the union of documents appearing in those score files for a query; if the union exceeds the dataset-specific pool size, it is truncated by RRF with $k = 60$ over the full native ranked lists, following the conventional RRF constant used by Cormack et al. and common retrieval toolkits such as Anserini/BEIR fusion utilities [7]. The study therefore evaluates a specific *RRF-centered candidate-pool setting*. It does not claim candidate-pool neutrality across alternative pool construction policies.

The study uses exactly the same stored score files, candidate pools, and qrels for all protocols. Differences among `ms2`, `ms1`, and `ms1_drop_mutual` therefore reflect only vote extraction and graph construction, not changes in upstream evidence.

4.3 Methods and Baselines

The normalized-protocol package evaluates four graph-independent baselines and ten graph-dependent methods. The graph-independent baselines are Prior, RRF, CombSUM, and Borda fusion. The graph-dependent methods are unrepaired and repaired variants of Copeland, balance, and Markov extraction, plus unrepaired and repaired Copeland and balance hybrids at $\alpha = 0.3$.

CombSUM uses per-query, per-ranker min–max normalization before summation, so missing documents contribute 0 for that ranker. Borda fusion operates on the rank positions of the stored score lists over the union of scored documents for that query. Prior and the RRF baseline both use the RRF family with $k = 60$ on candidate-restricted lists, but they are not bitwise-identical implementations: Prior ranks by fused prior score with document-id tie-breaking, whereas the RRF baseline breaks ties by best observed rank and then document id. That tie-breaking difference alone yields identical full rankings in only 216 of 6,156 comparable Prior–RRF cases, where 6,156 is every query-regime graph evaluated anywhere in the full study — 342 usable queries (Table 2) $\times$ 3 vote regimes $\times$ 6 protocol variants (the primary normalized protocol, the raw-margin ablation, and four further normalization/threshold variants used for the sensitivity checks in Section 9), i.e. $342 \times 3 \times 6 = 6{,}156$. We therefore report both and treat them as related but non-interchangeable baselines. These classical fusion baselines serve as graph-independent controls for the repair-sensitivity question; the study does not aim to establish absolute retrieval leadership against modern cross-encoder or LLM rerankers, which would require a different experimental design with new inference runs and are outside the measurement contribution pursued here.

The graph-independent baselines are identical across vote regimes for a fixed query because they operate directly on the stored score files rather than on the constructed preference graph. Regime-labeled baseline rows are reported only to align comparisons with graph-dependent methods; they are not independent reruns.

Table 3. Method families evaluated in the normalized four-dataset package.

Method family	Uses graph?	Repaired and unrepaired?
Prior	No	—
Reciprocal rank fusion (RRF)	No	—
CombSUM	No	—
Borda-count fusion	No	—
Copeland graph	Yes	Yes
Balance graph	Yes	Yes
Markov graph	Yes	Yes
Copeland hybrid (Eq. (15))	Yes	Yes
Balance hybrid (Eq. (15))	Yes	Yes

4.4 Repair Configuration

All primary results use the greedy cycle-peeling repair described in Section 3.8. To check whether this heuristic's known sub-optimality changes either the structural or the retrieval conclusions, we additionally solved the exact minimum-weight feedback arc set problem (Eq. (9)) with an open-source mixed-integer-programming solver (SCIP via PySCIPOpt 6.2.1; no commercial solver dependency) and compared it, method by method and on the same graph, against greedy repair.

Scope: 1,025 of 1,026 query-regime graphs. The 342 usable queries in Table 2, each evaluated under 3 vote regimes, give $342 \times 3 = 1{,}026$ query-regime graphs under the primary protocol. This check covers 1,025 of them: exactly one cell (BRIGHT, regime `ms2`, query `biology:0`) retains zero votes from every ranker under that regime's thresholds, so its preference graph has zero edges. An edgeless graph has no feedback arc set to compute — greedy and exact repair both trivially remove nothing — so we exclude it under a fixed rule applied uniformly to this check: a query-regime graph is included iff it retains at least one edge. This is the only excluded cell among the 1,026; it does not affect Table 2's usable-query counts, which count eligible queries, not retained edges. Every one of the 1,025 solves was accepted only under SCIP's proven-optimal status with MIP gap 0 (mean solve time 7.4 ms, none near the 300 s safety limit), and cross-checked against brute-force exact enumeration on 49 held-out graphs (34 synthetic graphs at four sizes and three densities, plus 15 real HotpotQA `ms1` graphs); all 49 objective values matched. Table 4 reports both configurations.

Structural comparison. Among the 379 of these 1,025 queries whose raw graph actually contains a cycle (repair is trivial on the remaining 646), the exact solver selects a different removed-edge set than greedy in 87.9% of them and removes 26.3% less total edge weight on average, confirming that greedy cycle peeling is measurably, not just theoretically, sub-optimal on this data.

Retrieval comparison. This structural difference does not change the retrieval conclusion. We compare greedy- versus exact-repaired nDCG@5, nDCG@10, nDCG@20, MRR, and MAP (paired per-query delta, exact minus greedy) for every repaired-graph-dependent ranking rule: the Copeland, balance, and Markov graph rankings and their $\alpha = 0.3$ hybrids (Section 3.9), plus topological and priority-topological ranking, two further repaired-only extraction rules with no unrepaired counterpart that are reported only in this check. That is 7 methods × 5 metrics = 35 pooled cells over all 1,025 queries. Split by the 12 dataset×regime cells this is $9 \times (7 \times 5) + 3 \times (7 \times 4) = 399$: HotpotQA's candidate pool has only $|D_q| = 10$ documents (Table 2), so nDCG@20 is undefined there and each of its 3 regimes contributes 28 rather than 35 cells. (nDCG@5 is included for every dataset as an additional robustness cutoff for this repair-algorithm

Table 4. Repair configurations used in the primary analysis and robustness check.

Configuration	Role
Greedy cycle peeling	Primary repair for normalized and raw-ablation runs, all 1,025 queries
Exact MWFAS (SCIP / PySCIPOpt 6.2.1)	Robustness check only, same 1,025 queries; confirms greedy's structural sub-optimality but no change to the retrieval conclusion

check, not as a substitute for the dataset-specific primary cutoff used elsewhere in the paper.) Holm and Benjamini–Hochberg correction is applied twice, as two separate joint families — once over the 35 pooled cells, once over the 399 per-dataset-regime cells — and no cell in either family survives Holm correction. We therefore report the exact solver only as a robustness check on the retrieval conclusion, not as a second primary repair configuration, and continue to use greedy cycle peeling for all primary nDCG analyses.

4.5 Statistical Analysis

All repaired-versus-unrepaired comparisons are paired at the query level. For each cell we report the mean delta, median delta, 5th/25th/75th/95th-percentile query-level deltas, and helped/harmed/unchanged query counts. We report a paired sign-flip permutation test and a paired percentile bootstrap interval with 10,000 resamples, treating significance, interval uncertainty, and multiplicity-corrected decisions as distinct layers of evidence [26]. If δ_i denotes the per-query repaired minus unrepaired nDCG difference for a paired set of n queries, and $\bar{\delta}^{(b)}$ is the mean of bootstrap resample b, the reported percentile interval is

$$\left[\bar{\delta}^{(\lceil 0.025(B-1)\rceil)},\ \bar{\delta}^{(\lceil 0.975(B-1)\rceil)}\right], \tag{17}$$

with $B = 10{,}000$. Bootstrap summaries additionally include the fraction of bootstrap means greater than zero.

The primary inferential family is the set of repaired-versus-unrepaired nDCG cells under the min–max plus retention-matched protocol. We apply both Holm and Benjamini–Hochberg correction across that family. We do not treat a bootstrap interval endpoint equal to zero as evidence of a strictly positive effect, and we do not infer robustness from percentile intervals alone. For interpretation-sensitive cells we additionally report leave-one-query-out and top-influence-removal analyses.

As a secondary robustness check, we recompute MAP and mean reciprocal rank (MRR) from the same stored candidate scores, pools, and rankings used for nDCG, without regenerating upstream retrieval. Those binary-relevance metrics are exploratory relative to the primary nDCG family: they are not folded into the main Holm/BH correction, and pool-truncated Recall@k is omitted because candidates are already restricted before ranking. Across MAP and MRR, no repaired-versus-unrepaired cell survives Holm correction; a small mixed-sign BH subset remains (especially SciDocs `ms1` positive and BRIGHT `ms1` negative), so the secondary metrics do not overturn the primary nDCG conclusion.

4.6 Implementation and Reproducibility

The primary analysis pipeline is implemented in Python 3.12.3 using `networkx` for graph operations. Run manifests record the executable, stored score-file hashes, qrels hashes, query-ID lists, protocol definitions, threshold tables, and generated outputs. Because this is an anonymous review draft and no verified anonymized artifact URL was available at writing time, we describe the artifact contents here rather than inserting a public author-identifying repository link.

The primary analysis is deterministic given the stored inputs and recorded seeds. In particular, it reuses fixed query-ID manifests instead of redrawing query samples. Bootstrap intervals use seed 13 with $B = 10{,}000$ resamples; paired sign-flip permutation tests use Monte Carlo sampling with 10,000 replications and seed 17 (not an exact 2^n enumeration). The study does not claim exact reproducibility for separate real-LLM judgment pipelines.

4.7 Real-LLM Evidence: Scope and Role

In addition to the mechanical three-ranker evaluation described above, we inspect stored real-LLM judgment corpora on subsets of SciDocs, HotpotQA, and FiQA, but we no longer treat those records as a parallel retrieval evaluation. Section 7.2 reframes them as a bounded protocol-quality check because the stored corpora differ from the mechanical study in prompt symmetry, parser behavior, exclusion logging, and provider coverage. We flag that distinction here because those corpora are easy to conflate with the main mechanical-vote evaluation if their weaker evidentiary status is not made explicit.

5 STRUCTURAL DATA QUALITY RESULTS

This section reports the primary normalized protocol introduced in Section 3.2: per-query, per-ranker min–max score normalization followed by retention-matched vote thresholds. The unnormalized raw-margin construction is retained only as an ablation for sensitivity analysis. We separate four questions that a raw-margin analysis alone conflates: score-scale imbalance, graph construction, the share of cyclicity attributable to direct mutual contradictions, and the amount of structural work repair actually performs.

5.1 Raw Score Scale Imbalance

The raw protocol is strongly dominated by native BM25 scale. Under the unnormalized raw-margin construction, BM25 contributes a mean conditional edge weight share of 0.988 whenever it participates in an edge; under the primary protocol, the corresponding mean falls to 0.512. This shift does not show that BM25 becomes less useful after normalization. It shows that the unnormalized weighted graph was largely tracking one ranker's score scale, not combining commensurate evidence from BM25, TF-IDF, and MiniLM. A supporting analysis of the fixed raw thresholds further shows they were not semantically comparable across rankers, so raw score margins cannot be interpreted as cross-ranker confidence weights. Figure 2 shows this shift directly.

5.2 Normalization Changes Graph Construction

Normalization changes more than the numeric scale of existing edges. It changes which edges are retained, how much total weight they carry, and which edges are ultimately selected for repair. The overlap between the primary protocol's removed edges and the raw-margin ablation is only 0.090 on SciDocs `ms1`, 0.055 on FiQA `ms1`, 0.328 on HotpotQA `ms1`, and 0.110 on BRIGHT `ms1`. These low overlaps show that normalization changes repair decisions themselves, not merely the magnitudes assigned to an otherwise stable set of edges. Under the primary protocol, `ms2` remains acyclic on all four datasets, while `ms1` is still highly cyclic before mutual-pair deletion. Figure 3 summarizes that primary regime-level pattern and Figure 4 contrasts it with the raw-margin ablation.

5.3 Mutual Pairs Versus Residual Nontrivial Cyclicity

The normalized graphs remain cyclic under `ms1`, but a substantial share of that cyclicity comes from direct mutual contradictions rather than longer preference loops. Averaged over all dataset×regime cells in the primary protocol,

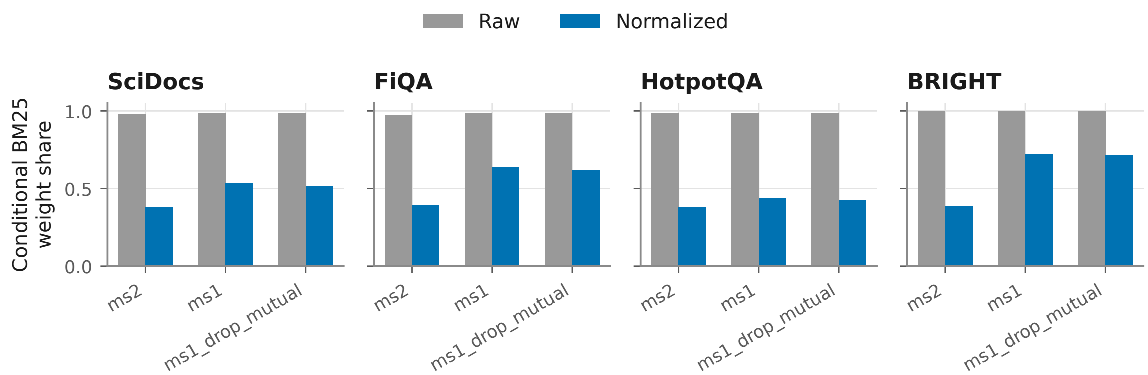

Fig. 2. Mean conditional BM25 edge-weight share by dataset, vote regime, and protocol: gray bars are the raw-margin ablation, blue bars are the primary normalized protocol (per-query/per-ranker min–max normalization with retention-matched thresholds). BM25 share is near 1.0 under the raw protocol in every panel and drops to roughly 0.4–0.7 after normalization, confirming that the raw graph was tracking BM25's score scale rather than combining three rankers' evidence. This is a graph-construction diagnostic, not a retrieval-quality measurement.

Table 5. Intrinsic graph diagnostics under the primary normalized protocol. Cyclic% is the share of query graphs containing at least one directed cycle before mutual-pair deletion; Post-mutual Cyclic% recomputes the same statistic after deleting direct mutual pairs; Largest SCC is the mean size of the largest strongly connected component before deletion; normalized FAS is mean removed repair weight divided by total graph weight. Raw-margin results are treated only as an ablation and are not shown as the main evidence here.

Dataset	Regime	Cyclic%	Post-mutual Cyclic%	Largest SCC	Norm. FAS
SciDocs	ms2	0.0	0.0	1.00	0.000
SciDocs	ms1	99.2	10.8	16.17	0.074
SciDocs	ms1_drop_mutual	11.7	11.7	1.93	0.002
FiQA	ms2	0.0	0.0	1.00	0.000
FiQA	ms1	98.3	30.8	16.12	0.080
FiQA	ms1_drop_mutual	30.8	30.8	3.38	0.004
HotpotQA	ms2	0.0	0.0	1.00	0.000
HotpotQA	ms1	63.5	1.9	4.06	0.029
HotpotQA	ms1_drop_mutual	1.9	1.9	1.13	0.000
BRIGHT	ms2	0.0	0.0	1.00	0.000
BRIGHT	ms1	92.0	22.0	10.82	0.055
BRIGHT	ms1_drop_mutual	22.0	22.0	3.08	0.003

cyclic-query prevalence is 0.3495 before mutual-pair deletion and 0.1100 afterward. The contrast is especially sharp on HotpotQA: its `ms1` cyclic-query rate drops from 63.5% to 1.9% after direct mutual pairs are deleted. SciDocs, FiQA, and BRIGHT retain more residual nontrivial cyclicity after deletion (10.8%, 30.8%, and 22.0%, respectively), but the residual graphs are structurally much smaller than the full `ms1` graphs. Figure 5 shows this before/after decomposition directly.

5.4 Structural Repair Activity

Repair is active almost entirely where cyclicity remains active. Under the primary protocol, `ms2` removes zero weight on all four datasets, while `ms1` removes nontrivial normalized weight on every dataset and `ms1_drop_mutual` removes only a small residual amount. Figure 6 reports normalized removed weight directly. As a secondary diagnostic, Table 6 reports qrels-anchored BEW/PIC reductions under the normalized `ms1` graphs. These quantities are computed

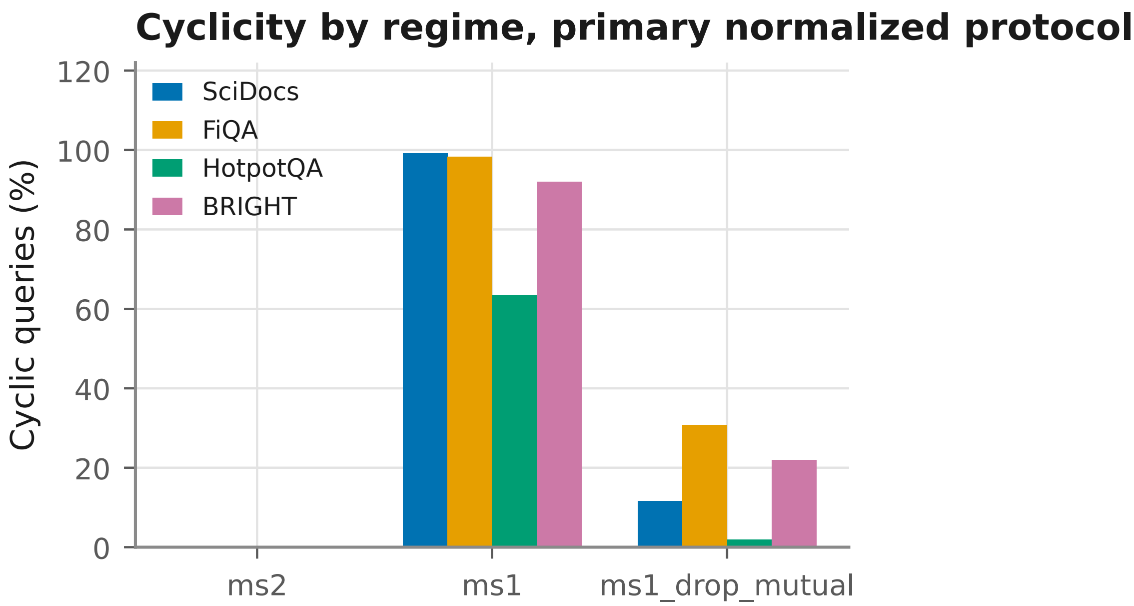

Fig. 3. Cyclic-query percentage under the primary normalized protocol, grouped by vote regime with one bar per dataset. `ms2` is exactly 0% on every dataset; `ms1` is 64–99%; deleting direct mutual pairs (`ms1_drop_mutual`) brings every dataset back down to 2–31%. The reader should notice that `ms1`'s high cyclicity is concentrated almost entirely in direct mutual contradictions, not longer cycles – Figure 5 isolates that residual directly. Values before mutual-pair deletion; post-deletion values are also in Table 5.

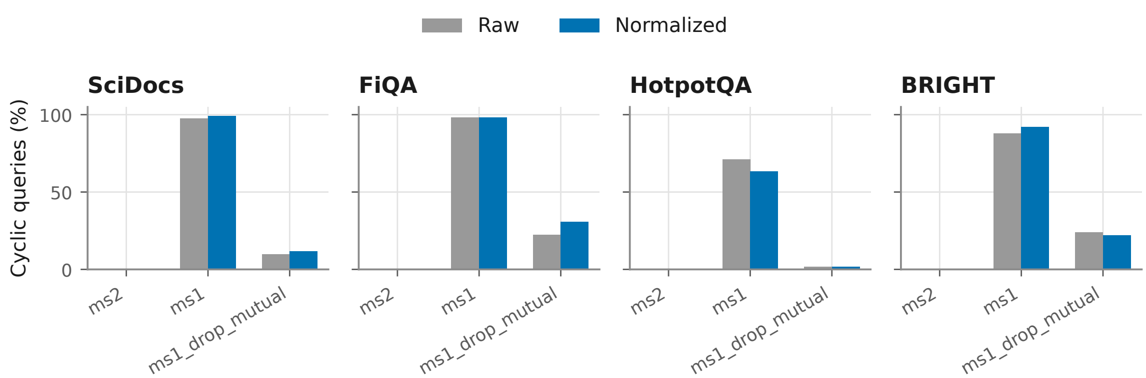

Fig. 4. Cyclic-query percentage by vote regime, raw-margin ablation (gray) versus primary normalized protocol (blue), one panel per dataset. Unlike the BM25 weight share in Figure 2, overall cyclic-query rate changes only modestly between raw and normalized construction within each regime. The reader should not conclude that normalization leaves the graph unchanged: Section 5 shows the removed-edge overlap between raw and normalized repair is only 0.055–0.328, so normalization changes *which* edges are cyclic and removed far more than it changes the headline cyclic-query rate shown here.

against the same relevance-derived reference used later for nDCG@k evaluation, so they should be read as supporting diagnostics rather than independent evidence that structural repair must improve retrieval.

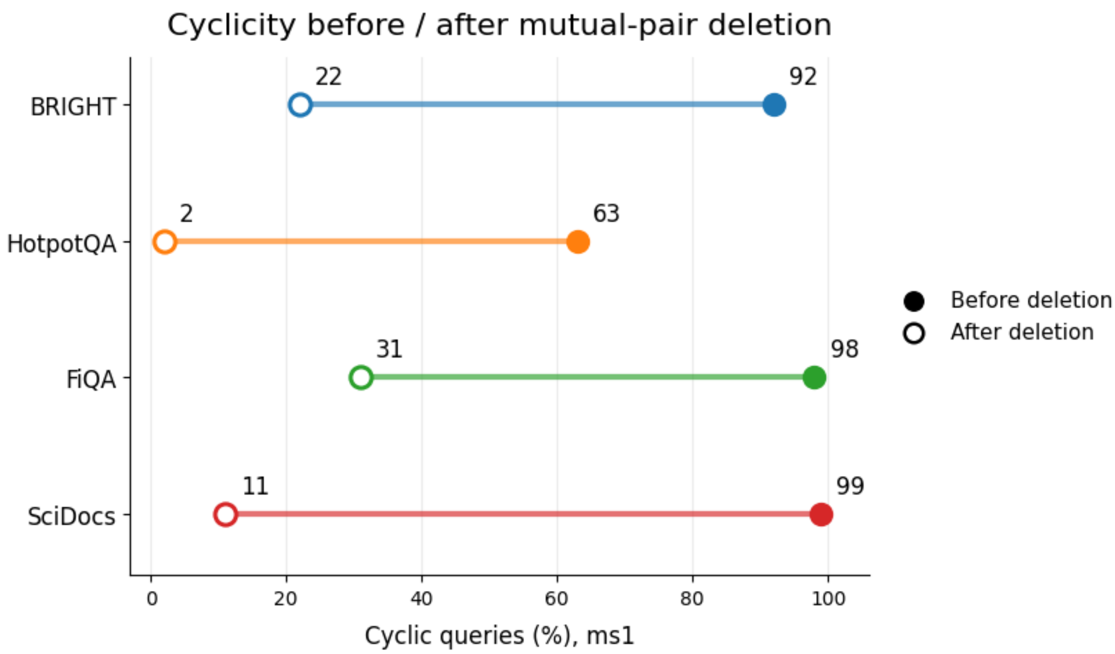

Fig. 5. Cyclic-query percentage under ms1 before (filled dot) and after (open dot) deleting direct mutual pairs, one row per dataset. The drop is largest for HotpotQA (63% $\rightarrow$ 2%) and smallest for FiQA (98% $\rightarrow$ 31%). The reader should take away that most ms1 cyclicity is direct bidirectional contradiction, not longer-cycle nontransitivity, and that the remaining share after deletion varies substantially by dataset.

Table 6. Qrels-anchored structural diagnostics under the primary normalized protocol, ms1 only. ΔBEW and ΔPIC are mean pre-repair minus post-repair changes computed against the same relevance-derived reference ranking used for downstream nDCG evaluation; mean removed weight is the intrinsic repair objective. These diagnostics are therefore descriptive and qrels-anchored rather than independent validation.

Dataset	ΔBEW	ΔPIC	Weight removed
SciDocs	4.250	19.49	8.934
FiQA	3.919	18.24	7.974
HotpotQA	0.563	2.25	1.173
BRIGHT	2.108	9.36	4.370

6 DOWNSTREAM QUALITY RESULTS: REPAIR VS. RETRIEVAL

The normalized structural results above do not answer the retrieval question by themselves. This section therefore evaluates repaired versus unrepaired ranking quality directly under the primary normalized protocol, then asks whether any positive point estimates survive paired testing, multiplicity correction, and influence analysis. We next compare those normalized graph-based methods with simple fusion baselines, examine raw-versus-normalized sign instability, and close with the limited alpha sweep and balance-degeneracy analysis.

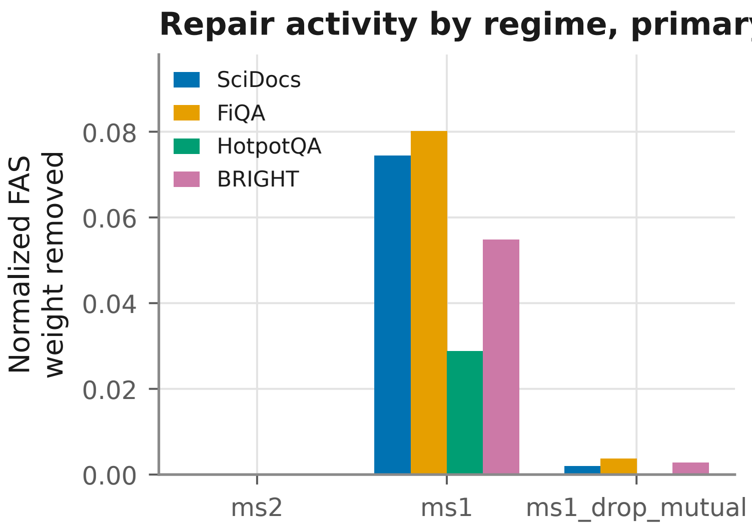

Fig. 6. Mean normalized FAS weight removed (removed weight divided by total graph weight) by vote regime, one bar per dataset, under the primary normalized protocol. ms2 removes exactly 0 on every dataset by construction (it is always acyclic); ms1 removes 0.029–0.080 of total graph weight; ms1_drop_mutual removes only a small residual (≤ 0.004). This is an intrinsic graph diagnostic computed before any retrieval evaluation, not a measure of retrieval benefit.

Table 7. Primary normalized repaired-minus-unrepaired nDCG@k for the active ms1 Copeland cells. Intervals are paired percentile bootstrap intervals over common queries; p_{perm} is the paired permutation test p-value on the same query set. No multiplicity adjustment is applied inside this table; corrected results are discussed in the next subsection.

Dataset	Pair	Mean Δ	95% CI	p_{perm}	+/−/=
SciDocs	Copeland graph	+0.011	[0.002, 0.022]	0.020	36/22/62
SciDocs	Copeland hybrid	+0.009	[0.002, 0.017]	0.012	30/10/80
FiQA	Copeland graph	+0.004	[−0.005, 0.011]	0.441	21/10/89
FiQA	Copeland hybrid	−0.005	[−0.013, 0.002]	0.301	16/8/96
HotpotQA	Copeland graph	+0.016	[−0.010, 0.044]	0.314	5/3/44
HotpotQA	Copeland hybrid	+0.012	[0.000, 0.031]	0.253	3/0/49
BRIGHT	Copeland graph	−0.014	[−0.040, 0.006]	0.267	19/13/18
BRIGHT	Copeland hybrid	+0.001	[−0.016, 0.020]	0.910	10/10/30

6.1 Repaired vs. Unrepaired Retrieval Under the Primary Protocol

Table 7 reports the active ms1 cells for the normalized Copeland graph and Copeland hybrid. Figure 7 plots the complete normalized primary grid, including the many zero-effect ms2 and ms1_drop_mutual comparisons. Positive point estimates exist in several cells, but none remains robust after the full testing and sensitivity pipeline reported below.

Most normalized repaired-versus-unrepaired cells are exactly zero because ms2 is acyclic and ms1_drop_mutual is nearly acyclic on most queries. The substantive activity is concentrated in ms1, but even there the signs are mixed across datasets and methods. SciDocs ms1 is the strongest positive point-estimate case for Copeland, with paired mean deltas of +0.011 for the graph ranking and +0.009 for the hybrid. FiQA ms1 shows a positive Copeland graph estimate but a negative Copeland hybrid estimate. BRIGHT ms1 is near zero for the Copeland hybrid and negative for the Copeland graph. HotpotQA ms1 retains positive Copeland point estimates, but the effect is influence-sensitive and

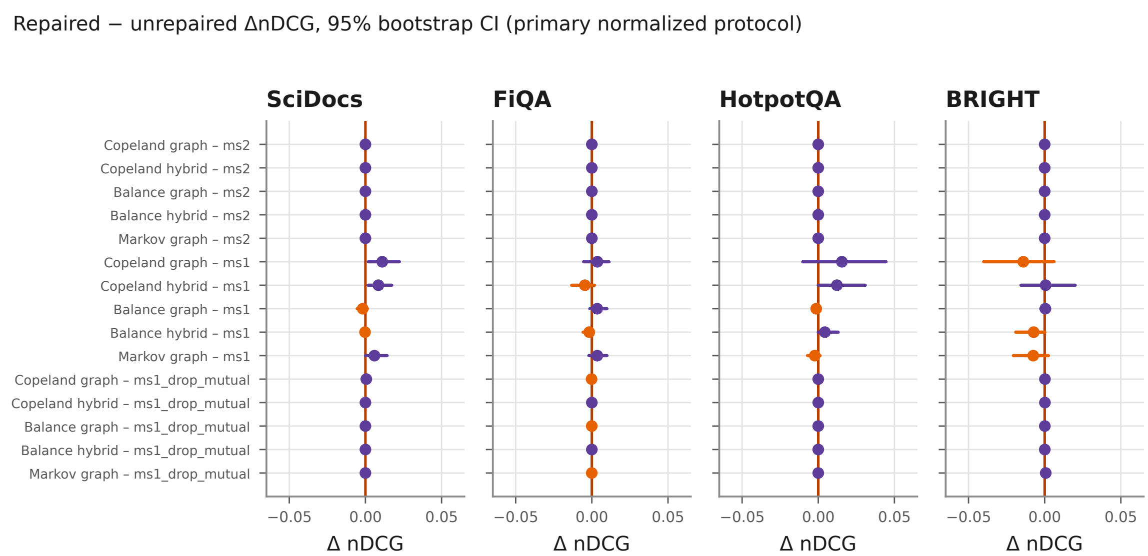

Fig. 7. Repaired-minus-unrepaired ΔnDCG with paired 95% bootstrap intervals, one panel per dataset and one row per regime×method pair (15 rows: 5 method pairs × 3 regimes), with a bold vertical line at zero. Every `ms2` and `ms1_drop_mutual` row collapses to a point exactly at zero; only `ms1` rows show visible intervals, and most of those still cross zero. The reader should notice that the handful of intervals not touching zero (e.g., SciDocs and HotpotQA Copeland rows) are exactly the cells discussed in the text below – and that even they do not survive multiplicity correction (Section 6).

does not survive the robustness checks below. The HotpotQA Markov graph also changes sign relative to the raw-margin ablation, moving from a positive mean in the raw protocol to a negative mean (−0.0023) under the normalized primary protocol.

6.2 Multiplicity and Influence Robustness

None of the positive repaired-versus-unrepaired cells remains robust after the primary multiplicity correction family is applied. The two smallest raw paired permutation p-values occur on SciDocs `ms1` Copeland hybrid ($p = 0.012$) and SciDocs `ms1` Copeland graph ($p = 0.020$), but both lose support after Holm and Benjamini–Hochberg adjustment. The HotpotQA `ms1` Copeland hybrid, which previously supplied the paper's positive headline, has a fresh paired permutation p-value of approximately 0.253 under the primary normalized protocol and an interval whose lower endpoint is exactly zero rather than strictly positive.

Figure 8 plots this influence pattern directly for the more concentrated HotpotQA cell. HotpotQA's positive mean is concentrated in a handful of queries. Removing the top one influential query reduces the normalized `ms1` Copeland-hybrid mean delta from +0.0123 to +0.0053; removing the top two reduces it to +0.0009; removing the top three or top four drives the mean to exactly zero. SciDocs is less concentrated: the corresponding Copeland-hybrid mean remains positive after removing the top one, two, three, and four influential queries (+0.0055, +0.0036, +0.0048, and +0.0041), but these nominal positives still do not survive multiplicity correction. The primary evidence therefore supports influence-sensitive positive point estimates in a small number of cells, not a robust positive retrieval effect of repair.

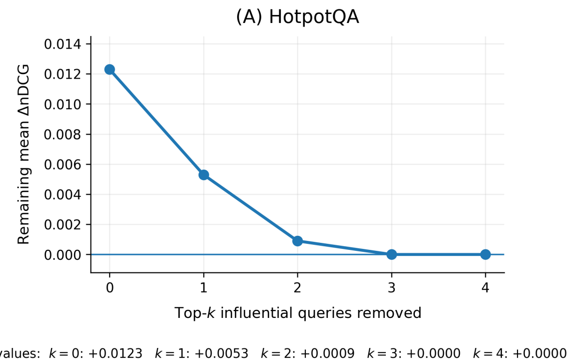

Fig. 8. Remaining mean ΔnDCG for the normalized `ms1` Copeland-hybrid cell as the top-k most influential HotpotQA queries are removed ($k = 0, \dots, 4$), the more concentrated of the two main positive-point-estimate cases. The mean falls from +0.0123 to exactly 0 after removing the top 3 queries, so the positive estimate is concentrated in a handful of queries. The companion SciDocs cell is less concentrated and is reported numerically in the text below rather than plotted here. Values are exact per-query recomputations, not interpolated.

Table 8. Sign changes between the raw-margin ablation and the primary normalized protocol. The raw protocol uses native heterogeneous scores with fixed thresholds and is shown only as an ablation. The normalized protocol uses per-query/per-ranker min–max normalization with retention-matched thresholds. Values are repaired-minus-unrepaired mean nDCG@k.

Dataset	Cell	Raw	Normalized
SciDocs	`ms1` Copeland graph	−0.0079	+0.0110
SciDocs	`ms1` Copeland hybrid	−0.0018	+0.0085
FiQA	`ms1` Copeland graph	−0.0029	+0.0036
BRIGHT	`ms1` Copeland hybrid	−0.0045	+0.0006
HotpotQA	`ms1` Markov graph	+0.0087	−0.0023

6.3 Raw Versus Normalized Sign Instability

Normalization changes retrieval conclusions as well as structure. Table 8 lists the five most consequential sign flips between the raw-margin ablation and the primary normalized protocol. These flips are not incidental numeric perturbations: they accompany large shifts in edge retention, edge weighting, and removed-edge overlap, so they show that the paper's downstream conclusions are not normalization invariant. Figure 9 plots the full raw-versus-normalized comparison underlying Table 8.

6.4 Comparison with Fixed Aggregation Baselines

The normalized repaired-versus-unrepaired analysis isolates the marginal effect of repair, but it does not answer whether graph-based methods outperform simple fusion baselines. Table 9 therefore reports dataset macro-averages under the primary `ms1` protocol, and Figure 10 shows the corresponding per-dataset baseline comparison. We use dataset-level summaries rather than pooled query×regime means because the normalized conclusions are not uniform across datasets.

Simple fusion baselines remain strong after normalization. CombSUM has the highest dataset-macro mean under primary `ms1`; RRF is also competitive, and both outperform the repaired Copeland graph methods on this summary.

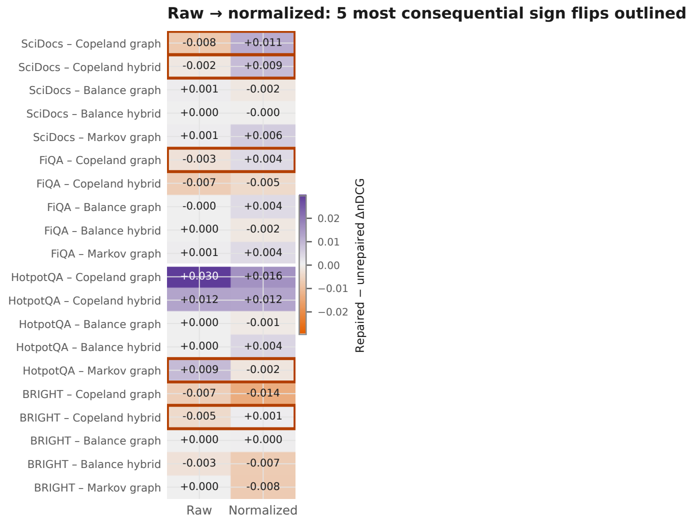

Fig. 9. Repaired-minus-unrepaired ΔnDCG for every dataset×method pair under `ms1`, raw-margin ablation versus primary normalized protocol, colored by sign and magnitude (purple positive, orange negative). The five rows outlined are the sign flips Table 8 calls out in the text; most other cells keep the same sign, or flip only at noise-level magnitude on both sides. The reader should take away that a handful of specific, named conclusions reverse under normalization – not that every cell is unstable.

Table 9. Dataset-macro comparison under the primary normalized protocol, `ms1` only. Mean nDCG is averaged across the four datasets; average rank is the mean within-dataset rank by mean nDCG. Because Prior and RRF use closely related but not identical implementations, both are reported separately rather than treated as the same baseline.

Method	Macro mean nDCG	Avg. rank
CombSUM	**0.554**	3.25
Balance hybrid unrepaired	0.548	4.75
Balance hybrid repaired	0.547	6.00
RRF	0.546	5.50
Copeland hybrid repaired	0.546	6.50
Prior	0.545	6.75
Borda fusion	0.545	6.00
Copeland hybrid unrepaired	0.542	9.00
Balance repaired	0.539	8.25
Balance unrepaired	0.539	8.25
Markov unrepaired	0.537	9.50
Markov repaired	0.537	9.25
Copeland repaired	0.534	11.00
Copeland unrepaired	0.530	11.00

The Prior baseline uses the same RRF family on candidate nodes but differs from the standalone RRF baseline in implementation details and exact tie-breaking, so we report both rather than treating them as interchangeable; as noted in Section 4.3, their exact rankings match in only 216 of 6,156 comparable cases across the full study. The primary evidence therefore does not support a claim that graph repair makes simple baselines obsolete.

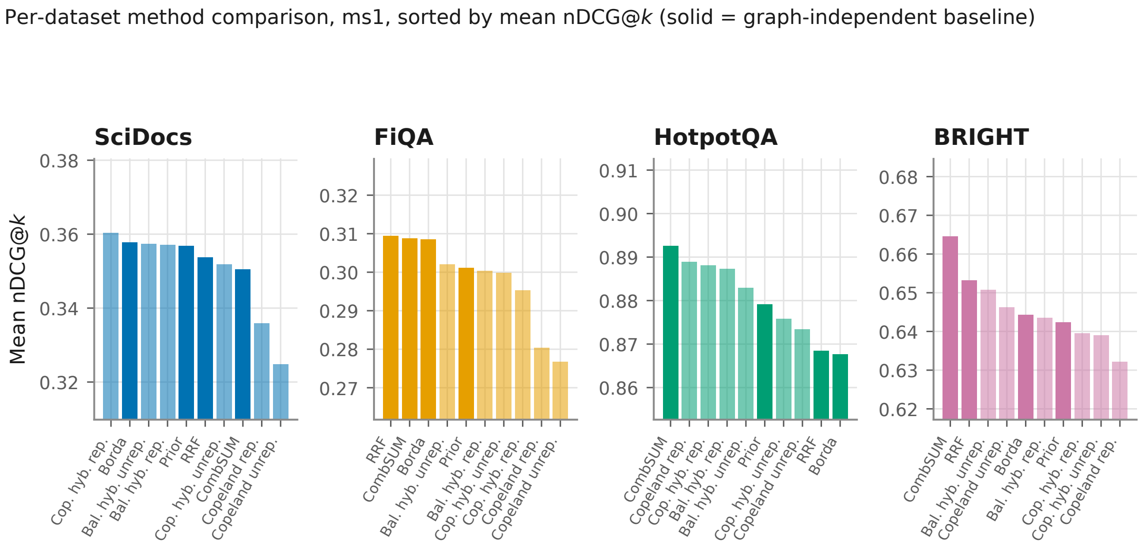

Fig. 10. Mean nDCG@k under `ms1`, one panel per dataset, methods sorted left to right by value; solid bars are the four graph-independent baselines (Prior, RRF, CombSUM, Borda), translucent bars are graph-dependent methods. Note the per-dataset y-axis: each panel is scaled to its own score range so within-dataset ordering is visible, so only within-panel comparisons are meaningful, not bar heights across panels. A graph-independent baseline is within the top three methods on every dataset (CombSUM leads on HotpotQA and BRIGHT, RRF on FiQA), while on SciDocs a repaired Copeland-hybrid edges out CombSUM by a narrow margin – consistent with Table 9's macro-average ranking, which pools across all four datasets rather than reporting any single one.

Table 10. Copeland-hybrid alpha sensitivity under the primary normalized protocol, `ms1` only. Entries are repaired-minus-unrepaired mean nDCG@k; this is a small sensitivity sweep, not a tuned hyperparameter search.

Dataset	$\alpha = 0.1$	$\alpha = 0.3$	$\alpha = 0.5$	$\alpha = 1.0$
SciDocs	−0.0022	+0.0085	+0.0097	+0.0083
FiQA	+0.0006	−0.0046	+0.0010	+0.0038
HotpotQA	−0.0008	+0.0123	+0.0120	+0.0101
BRIGHT	−0.0013	+0.0006	−0.0074	−0.0071

6.5 Alpha Sensitivity and Balance Degeneracy

The small alpha sweep does not rescue a robust positive repair result. Table 10 reports the primary `ms1` Copeland-hybrid delta over $\alpha \in \{0.1, 0.3, 0.5, 1.0\}$. SciDocs remains positive for $\alpha \in \{0.3, 0.5, 1.0\}$, but those gains are still not robust after multiplicity correction. HotpotQA stays positive for $\alpha \in \{0.3, 0.5, 1.0\}$, but the effect remains influence-sensitive. FiQA and BRIGHT change sign within this small sweep, showing that stronger graph weight does not induce a stable benefit. Figure 11 plots all four datasets across the full α sweep.

The balance degeneracy analysis explains why many balance results remain exactly or nearly zero despite nontrivial graph changes. Under the primary `ms1` protocol, the graph changes on 119/120 SciDocs queries, 118/120 FiQA queries, 33/52 HotpotQA queries, and 46/50 BRIGHT queries. Those changes do not consistently propagate to the evaluation metric: the full balance-hybrid ranking changes on only 68, 68, 7, and 20 queries respectively, and nDCG changes on only 8, 5, 1, and 8 queries. In other words, the graph can change, the balance scores can change, and even the top-k

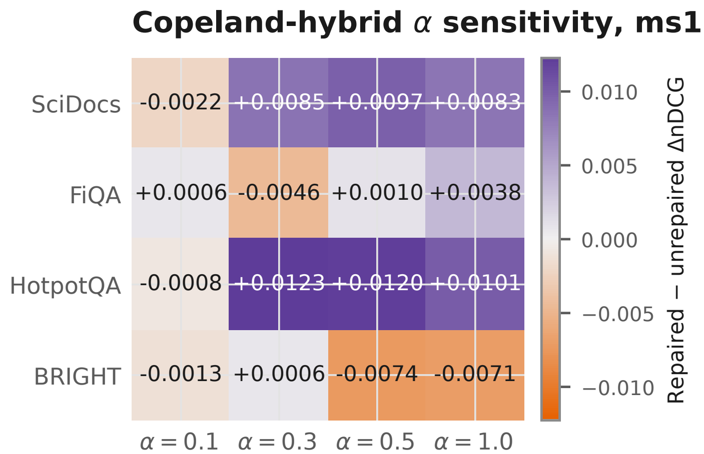

Fig. 11. Copeland-hybrid repaired-minus-unrepaired mean ΔnDCG under `ms1` as the graph-component weight α varies over $\{0.1, 0.3, 0.5, 1.0\}$, colored by sign and magnitude. SciDocs and HotpotQA stay positive for $\alpha \geq 0.3$; FiQA and BRIGHT change sign within this small sweep. The reader should take away that increasing the graph component's weight does not produce a consistent direction of effect across datasets; this is a small sensitivity sweep, not a tuned hyperparameter search or a multiplicity-adjusted test.

order can change without altering the relevance sequence scored by nDCG@k. The balance results are therefore not mysterious nulls; they are a direct consequence of downstream propagation failure from graph repair to the evaluated prefix.

7 SECONDARY ANALYSES AND SCOPE CHECKS

The normalized-protocol core evidence of this paper is contained in Sections 5 and 6. The four topics below are retained only as scoped secondary notes because they do not currently meet the same evidentiary standard as the normalized mechanical evaluation.

7.1 Rule-Based Outcome Taxonomy

A separate raw-margin failure-mining corpus of 1,020 records was previously labeled with a six-way rule-based outcome taxonomy. That taxonomy is not manually verified: it is an automated rule cascade whose final labels depend heavily on branch precedence, and its counts were never regenerated under the primary normalized protocol. Because normalization and vote construction change retained edges, removed edges, and repaired-versus-unrepaired signs, those raw-margin counts are not appropriate as evidence for the present study.

We therefore do not treat that taxonomy as part of the main empirical narrative. A *rule-based outcome taxonomy* may still be useful as future descriptive analysis, but it would need to be regenerated under the primary protocol before interpretation. Such a taxonomy would remain a retrospective outcome classification, not a validated predictive mechanism available before repair.

7.2 Protocol Audit of Stored LLM Judgments

We inspected stored real-LLM judgment corpora to determine whether they provide an independent validation of the normalized mechanical study. They do not currently support that role. The stored corpora exhibit parser defaults that silently map ambiguous or malformed outputs to a fixed label, measurable provider-dependent position bias, incomplete raw-text and exclusion logging, and protocol differences across providers and method families. These issues are serious enough that apparent cyclicity in a naively constructed LLM-derived graph can be dominated by pipeline artifacts rather than by genuine model disagreement.

Quantitatively, a stored pairwise-judgment corpus collected from the Cohere and Azure OpenAI APIs on FiQA, HotpotQA, and BRIGHT shows the following. The shared response parser silently defaults ambiguous or malformed outputs to a fixed label on 1.1% of 12,020 responses with preserved raw text (0.0% on FiQA for both providers; up to 3.8% on HotpotQA for Cohere). Among well-formed, unambiguous responses, both providers show statistically significant position bias, but in *opposite* directions: Azure prefers the first-shown candidate 53–58% of the time across the three datasets, while Cohere prefers the second-shown candidate 61–70% of the time (exact binomial $p < 0.01$ in every provider–dataset cell); pooling the two providers would largely cancel this out and hide both effects. Forward and reverse presentations of the same candidate pair agree with each other only 59–85% of the time depending on provider and dataset (Cohen's κ 0.27–0.71, fair to substantial). In a preference graph built directly from these judgments (not the mechanical graph used elsewhere in this paper), removing edges produced by the parser's default-label rule and by the position-bias-driven disagreement default reduces measured cyclicity by 91–100% (from a 31–62% cyclic range down to 0–5%), indicating that most of that graph's apparent cyclicity reflects these pipeline defaults rather than genuine model intransitivity. None of these figures alter the mechanical-graph statistics reported in Sections 5 and 6, which do not depend on any LLM judgment; they bound what a future LLM-judgment-derived graph-construction study would need to resolve before its cyclicity could be interpreted at face value.

For that reason, we do not pool the stored LLM corpora with the primary mechanical-vote evaluation or treat them as a parallel replication. The main lesson from these records is methodological: pairwise-judgment pipelines can manufacture apparent inconsistency if prompt symmetry, abstention behavior, parser handling, and exclusion accounting are not controlled carefully. Accordingly, we retain this material only as a bounded protocol-quality check.

A future LLM-based study intended to support the paper's central claims would need randomized candidate order, symmetric A/B and B/A prompting, abstention on malformed or ambiguous outputs, no default-label fallback, complete exclusion logs, and matched query/method protocols across providers.

7.3 Implementation Cost Note

We retain only a bounded implementation-cost observation. The normalized-protocol pipeline is CPU-only and practical on the graph sizes studied here, but the available runtime and memory evidence comes from a single unbenchmarked local machine with coarse resident-set-size snapshots and some timings near the instrumentation floor. We therefore do not present efficiency as a substantive contribution or make cross-system scaling claims. Repair cost depends on graph size, density, and SCC structure, and a controlled hardware benchmark would be required before stronger conclusions could be justified.

7.4 Planned CARB Resource

CARB is a planned feature-only companion resource, not a released benchmark in this review draft. No empirical claim depends on it. Any future release would need a review-ready package, corrected scope statistics, and licensing review consistent with the source datasets' redistribution terms.

8 DISCUSSION

Sections 5–7 support a conclusion more specific than a naive "repair either helps or it does not" framing: preference-graph repair cannot be interpreted independently of the score-normalization and vote-construction protocol that creates the graph. Under the primary normalized retention-matched protocol, repair still changes graph structure, but its downstream retrieval effect remains construction-sensitive, method-dependent, and not robustly positive across the tested benchmarks.

Two results were not anticipated at the outset of this study and are worth stating directly. First, the magnitude of raw score-scale dominance was larger than a general awareness of "BM25 and neural scores are not comparable" would predict: BM25 accounted for essentially all conditional edge weight (0.988) whenever it participated in an unnormalized edge, which means the raw-margin graph was, in practice, a single-ranker graph wearing a multi-ranker label. Second, the apparent cyclicity of the normalized `ms1` graphs is misleading if read at face value: most of it is direct bidirectional contradiction rather than longer-cycle nontransitivity, and the repair effect with the strongest positive point estimate (HotpotQA) collapses to exactly zero once three influential queries are removed. Neither result required a new algorithm to surface — both follow directly from treating graph construction and influence diagnostics as first-class measurements rather than fixed preprocessing.

Normalization is part of the model, not preprocessing trivia. The preference graph inherits whatever score scale is used to construct pairwise margins. Under the unnormalized raw-margin construction, those margins were not comparable across BM25, TF-IDF, and MiniLM, and BM25 contributed almost all conditional edge weight when it participated. The primary normalized protocol substantially reduced that scale domination and, with it, changed which votes survived, which edges were retained, and which edges were removed during repair. The practical lesson is methodological: conclusions about graph repair are inseparable from the normalization and threshold semantics used upstream. We do not claim that per-query, per-ranker min–max normalization is universally optimal, only that it is the best-supported primary protocol in this study because it materially reduced score-scale distortion and enabled a controlled comparison against the raw ablation.

Structural success does not guarantee retrieval gain. FAS repair (Eq. (9)) optimizes a graph-internal objective, whereas nDCG evaluates ranked retrieval against qrels. That distinction matters empirically: the normalized-protocol experiments still show nonzero repair activity in the cyclic regimes, yet structural improvement does not imply a better final ranking. Depending on the query, extraction rule, and fusion stage, a repaired graph can be retrieval-inactive, evaluation-invisible, neutral, or harmful. The qrels-anchored diagnostics in Section 5 are useful for interpretation, but they should not be treated as independent validation of the graph objective itself. The balance-degeneracy analysis (Section 6) identifies one concrete mechanism for this gap: a repaired graph can change on the large majority of queries while the induced ranking changes on a much smaller subset, and nDCG@k changes on fewer still, because a repair edit only registers on the metric if it reorders documents within the evaluated prefix. Structural consistency and retrieval effectiveness remain related but mechanistically distinct empirical targets, not two views of the same underlying quantity.

Mutual contradictions and longer-cycle structure should be reported separately. The normalized-protocol results also change how cyclicity should be read. Apparent graph cyclicity is substantial under the primary protocol before mutual pair deletion, but much of it arises from direct bidirectional contradictions rather than from longer cyclic structure. After deleting mutual pairs, the mean cyclic-query rate drops markedly, leaving a smaller but still nonzero residual of nontrivial cyclicity. This does not make the remaining cycles unimportant, but it does mean that total `ms1` cyclicity should not be interpreted automatically as genuine Condorcet-style nontransitivity. Future work should report both direct mutual contradiction and post-deletion cyclicity by default.

Repair effects are method- and normalization-dependent. The same repaired graph does not propagate uniformly through Copeland, Markov, balance, and the hybrid variants. Some normalized cells change sign relative to the raw package, and the balance-family zero-delta pattern in this study reflects propagation failure through the ranking pipeline in this study, not a general impossibility result for balance-based repair. The hybrid results similarly show that fusion can either dampen noisy graph changes or suppress potentially meaningful ones, depending on how much signal survives normalization and how strongly the prior ranking dominates the final score. Positive point estimates therefore require more than a favorable mean: in this study, none remains robust once paired testing, influence analysis, and multiplicity correction are applied together. HotpotQA is the clearest example of a positive-looking but non-robust cell: its normalized `ms1` Copeland-hybrid mean is positive, yet the effect is concentrated in a few influential queries, the paired permutation test is not compelling, and the cell does not survive Holm or Benjamini–Hochberg adjustment.

Simple baselines remain important under normalized graphs. The present study also does not support graph repair as a superior default retrieval strategy. Reciprocal-rank fusion and CombSUM remain strong comparators under the normalized protocol, and the prior ranking continues to absorb much of the useful signal in the hybrids. A plausible reason is architectural rather than incidental: RRF and CombSUM aggregate every ranker's evidence for a document directly, without an intermediate discrete decision about which pairwise margins clear a retention threshold, whereas graph construction introduces exactly that decision at the vote-extraction stage (Section 3.3). Each additional discrete choice is a further place where construction sensitivity can enter the pipeline, which is consistent with Benham and Culpepper's observation that fusion methods carry their own query-level risk-reward trade-offs rather than uniform gains [2]. At the same time, these comparisons should be read with the candidate-pool design in mind: the experiments are centered on an RRF-family pooling scheme, and Prior and the RRF baseline, although related, are not interchangeable rows. The right takeaway is therefore cautious rather than triumphalist: simple fusion baselines remain strong under the normalized protocol, and graph-dependent methods must justify their added complexity on a paired, dataset-specific basis.

Implications for future preference-graph studies. Preference-graph experiments should report normalization, threshold interpretation, common-query pairing, mutual-pair prevalence, residual nontrivial cyclicity, repair objective, ranking-extraction method, influence diagnostics, and multiplicity correction as first-class design choices rather than hidden preprocessing details. Raw-score constructions can still be informative as ablations, but not as primary evidence when the underlying rankers operate on heterogeneous score scales. This paper therefore does not offer a validated predictive selector for when repair will help. It offers a narrower but more defensible conclusion: repair outcomes depend materially on the graph-construction process, and any future claim about repair effectiveness should make that dependence explicit.

These findings matter beyond this specific repair heuristic because the same construction-before-measurement pattern recurs anywhere heterogeneous scores are turned into a derived structure before an algorithm is evaluated on it: learned fusion, cross-encoder score blending, and graph-based reranking all inherit whatever comparability assumption their upstream scores encode. A field that evaluates such algorithms without first establishing that the input structure is itself well formed risks attributing to the algorithm what is actually a property of its input. Treating graph quality as a measurable, reportable variable, rather than an assumed precondition, is the methodological contribution this paper offers to that broader concern.

8.1 Practical Implications

Beyond its methodological conclusion, this study supports concrete practices for teams that build or evaluate multi-ranker retrieval systems.

Auditing retrieval pipelines for score-scale dominance. A pipeline that merges BM25, TF-IDF, and neural similarity scores into a single weighted signal can be silently dominated by whichever ranker has the largest native score range. The conditional BM25 edge-weight share reported in Section 5 (0.988 raw versus 0.512 normalized) is a concrete, checkable diagnostic: a team combining heterogeneous ranker scores into any weighted representation, graph-based or not, can compute the same conditional-share statistic on its own score files before trusting the combined signal.

Deciding whether graph repair is worth the engineering cost. Feedback-arc-set repair adds a processing stage — cycle detection, edge removal, and re-ranking — to a retrieval pipeline. The evidence here indicates that this cost does not reliably translate into a measurable nDCG improvement once normalization is handled explicitly and paired statistical testing is applied. A team considering repair as a production feature should budget for the same evaluation discipline used here (paired testing, influence analysis, multiplicity correction) before adopting it, rather than assuming that reduced cyclicity implies an improved ranking.

Quality assurance for preference-graph construction. Because normalization and vote-extraction choices materially change retained and removed edges (Section 5), preference-graph construction should be treated as a data-quality-sensitive pipeline stage with its own audit trail: recorded normalization method, vote thresholds, mutual-pair rate, and cyclicity before and after mutual-pair deletion. The mutual-pair/nontrivial-cycle decomposition introduced here (Section 3.3) gives a concrete audit statistic that distinguishes direct bidirectional contradictions from genuine nontransitive structure, which a single cyclicity percentage would conflate.

Benchmarking graph-derived rankings against simple fusion. Reciprocal rank fusion and CombSUM are inexpensive graph-free baselines that remained competitive with, and in several cases stronger than, repaired graph-based rankings under the normalized protocol (Section 6). An evaluation of any new graph-based ranking method should include these baselines by default: their absence can make a graph-based method appear more valuable than it is relative to low-cost alternatives.

Evaluation protocol design. The raw-versus-normalized ablation shows that a fixed threshold policy can support a different repaired-versus-unrepaired conclusion than a retention-matched policy applied to the same underlying data. Reporting normalization method, threshold-selection procedure, and vote-extraction regime as explicit protocol parameters, rather than implementation detail, lets other researchers determine whether a reported repair effect is a property of the data or of the construction pipeline that produced the graph.

9 LIMITATIONS

This study's limitations fall into several categories, detailed below.

Normalization remains a design choice, not a proven optimum. The primary protocol uses per-query, per-ranker min–max normalization with retention-matched thresholds because that combination substantially reduced score-scale domination and supported the most defensible controlled comparison in this study. It is not the only plausible normalization. Other transformations in the pilot changed graph structure and some downstream signs, so the present paper demonstrates construction sensitivity rather than proving that one normalization is universally best.

Candidate pooling and baseline roles are RRF-centered. The experiments use an RRF-family candidate-pool construction and then compare graph methods against Prior, RRF, CombSUM, Borda, and related variants within that setting. Because candidate selection, Prior, and the RRF baseline occupy different implementation roles, conclusions may differ under independently constructed pools or different prior rankings. The paper documents these distinctions, but it does not establish pool-invariant behavior.

Nontrivial cyclicity is smaller than headline cyclicity. Under the primary protocol, cyclicity after removing direct mutual pairs is materially lower than cyclicity before deletion. That means some regimes offer limited opportunity for repair to act on longer-cycle structure, and the paper does not establish how the same methods would behave on corpora with intrinsically high nontrivial cyclicity.

Retention-target sensitivity changes structure more than retrieval robustness. The retention-matching procedure (Section 3.3) is an experimental control for comparing raw and normalized graph construction at approximately matched retention, not a universally optimal threshold-selection method. A sensitivity sweep of alternative retention/threshold policies shows that graph density, cyclicity, and removed-edge patterns change materially across policies, and some repaired-versus-unrepaired cell signs flip. Across those policies, however, no positive repaired-versus-unrepaired nDCG cell survives Holm correction. We therefore retain raw-reference retention matching as the primary protocol and interpret structural diagnostics as policy-sensitive, while the main multiplicity-corrected retrieval conclusion remains robust under the tested alternatives.

The repair algorithm is heuristic and order-sensitive. Our main repair procedure is greedy cycle peeling with no approximation guarantee, and this is the only repair used to produce the primary evidence in Sections 5–6. As a robustness check (Section 4.4), we additionally solved the exact minimum-weight feedback arc set problem with an open-source ILP solver on all 1,025 query-regime graphs with at least one retained edge (Section 4.4): the exact solution differs from greedy's on most cyclic queries and removes materially less weight, but this does not change which repaired-versus-unrepaired retrieval cells survive multiplicity correction. In principle, cycle-discovery order, insertion order, and alternative optimization strategies could still produce different greedy-repaired graphs and therefore different downstream rankings; the exact-solver check bounds, but does not eliminate, this concern for the primary greedy results.

Statistical power and influence remain limiting factors. Several cells have modest query counts, and some positive-looking means are concentrated in a small number of influential queries. The paper addresses this with paired permutation tests, bootstrap intervals, leave-one-out analysis, influence removal, and multiplicity correction, but those diagnostics also reinforce the main caution: no positive repaired-versus-unrepaired effect remains robust after Holm or Benjamini–Hochberg correction.

Metric scope is narrower than graph scope. nDCG is the main retrieval metric, and its cutoff determines whether some ranking changes are visible at all. MAP and MRR are reported only as secondary robustness checks and are not part of the primary multiplicity-corrected nDCG family; pool-truncated Recall@k is not used as a confirmatory metric. The qrels themselves are incomplete and dataset-dependent, and BEW/PIC are computed from the same relevance evidence rather than from an independent source of graph truth. The paper therefore does not claim that improved structural consistency implies a general improvement in user utility beyond the measured retrieval setting.

The stored LLM records are limited as evidence. As Section 7.2 now makes explicit, parser defaults, position bias, incomplete logging, and protocol differences prevent the current LLM artifacts from supporting a clean claim about genuine model nontransitivity or from serving as independent validation of the normalized mechanical study.

External validity remains bounded. The primary evidence is drawn from four retrieval benchmarks, three mechanical rankers, and a specific set of vote regimes, extraction methods, and fusion choices. We do not claim that the same normalized conclusions will hold unchanged for other domains, ranker families, pairwise-preference sources, or evaluation metrics.

The evaluated methods do not include learned fusion, cross-encoder rerankers, or online evaluation. All graph-independent baselines here (Prior, RRF, CombSUM, Borda) are unlearned, fixed-form combination rules, and all graph-dependent methods extract a ranking from a mechanically constructed graph. We do not evaluate a learned fusion model trained to combine the same ranker scores, a cross-encoder or LLM-based reranker applied to the same candidate pools, or any online (interleaving or click-based) evaluation of repaired versus unrepaired rankings. The stored LLM material discussed above (Section 7.2) is a protocol-quality check, not a controlled LLM-as-judge experiment measuring repair effectiveness, so this study does not report an LLM-as-judge comparison either. Learned fusion in particular could absorb some of the scale-comparability problem studied here automatically, so the comparative standing of graph repair against a tuned learned baseline remains an open question this study does not resolve.

Preference graphs are derived artifacts, not ground truth. Every structural diagnostic in this paper — cyclicity, mutual-pair rate, SCC size, removed weight — describes a constructed object whose properties follow from upstream score files, normalization, and thresholding, not an independent signal about document relevance. Two pipelines that start from the same upstream retrieval runs can produce different graphs, and therefore different structural and retrieval conclusions, if their construction choices differ. This paper audits one such construction pipeline in detail; it does not establish that preference graphs in general are a reliable representation independent of how they are built.

External companion resources remain optional. CARB is not released; evaluate this manuscript on the primary evidence reported here (Section 11), not on a promised companion release.

10 CONCLUSION

This study examines not only whether preference-graph repair affects retrieval, but also whether that conclusion is stable under defensible choices for normalizing and aggregating heterogeneous ranking scores. The central finding is methodological. Raw heterogeneous ranker margins severely distorted weighted graph construction, producing near-total BM25 dominance whenever BM25 participated. Replacing those raw margins with per-query, per-ranker min–max normalization and retention-matched thresholds materially changed retained edges, cyclicity, removed edges, and several repaired-versus-unrepaired signs.

Under this primary normalized protocol, repair still improves internal graph consistency in the active regimes, but structural consistency and retrieval effectiveness do not move together reliably. Positive repaired-versus-unrepaired point estimates remain in some cells, yet none survives the combined scrutiny of paired testing, influence analysis, and multiplicity correction. The paper therefore does not support a construction-invariant claim that repair helps, nor does it support a single robust positive exception. It instead shows that repair outcomes are sensitive to normalization and vote construction, and that simple fusion baselines remain strong comparators under the normalized protocol.

The practical implication is modest but important: preference-graph repair should be evaluated together with the score normalization, threshold semantics, mutual-pair handling, and ranking-extraction choices that determine the graph on which repair operates. Future work may yet find settings where repair yields a robust retrieval gain, but this study shows that such a claim cannot be treated as independent of graph construction.

The larger point extends past feedback-arc-set repair specifically. Preference- graph *quality* — not only repair-algorithm quality — governs what any downstream method can extract from a multi-ranker system, and that quality is measurable rather than assumed: conditional edge-weight share, mutual-pair prevalence, post-deletion cyclicity, and removed-edge overlap under alternative constructions all served here as concrete, reportable diagnostics. The robustness evidence in this paper is deliberately layered — paired testing, bootstrap uncertainty, leave-one-out influence analysis, multiplicity correction, and a raw-versus-normalized ablation — because any one of those checks alone would have understated how sensitive repair conclusions are to upstream construction choices. As graph-based representations become more common in retrieval and ranking research, this measurement sensitivity is not a peculiarity of feedback-arc-set repair; it is a methodological caution that generalizes to any method built on a derived preference structure. Future work on graph-based retrieval should therefore report and audit graph construction with the same rigor typically reserved for the downstream ranking algorithm, rather than treating the graph as a neutral input.

11 DATA AVAILABILITY AND REPRODUCIBILITY

Data availability. SciDocs, FiQA, HotpotQA, and BRIGHT are publicly available from their original sources [6, 18, 23, 28]. This paper redistributes no raw third-party document collection. CARB remains planned only (Section 7.4).

Artifact availability. The pipeline in Sections 3–4 is packaged as a reproducible code-and-data artifact. This double-anonymous draft does not include an author-identifying public repository URL, and no verified anonymized mirror URL is claimed here. A scrubbed anonymous artifact should be attached through the venue submission system before review (not by email request). The package contains stored ranker scores, query-ID manifests, qrels/score hashes, primary normalized and raw-ablation outputs, tables/figures, and commit manifests. The primary evidence set is the normalized retention-matched package; the raw-margin package is an ablation only.

Reproducibility. Mechanical results in Sections 5–6 regenerate from stored intermediates with the seeds, solver settings, and eligibility rule in Section 4. Upstream retrieval and paid APIs need not be rerun. Stored LLM records support protocol-quality checks only. The review artifact includes a dependency list (`requirements.txt`) together with the code snapshot used for mechanical regeneration.

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